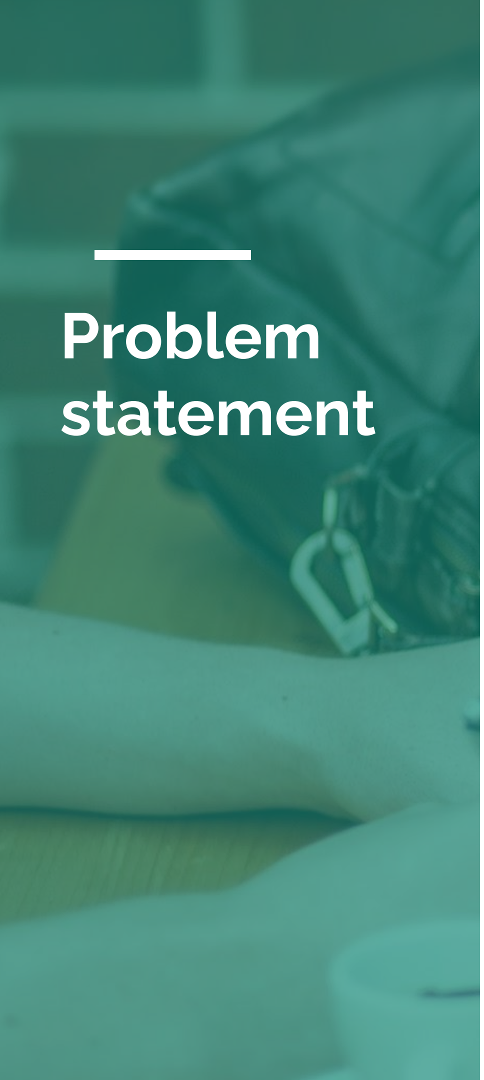
International Debt Data Analytics

An End-to-End Data Engineering & Visual Analytics Solution

To transform complex World Bank debt data into a structured relational database and an interactive dashboard.

Analyze debt distribution across countries and indicators.



Problem statement

SOURCE: World Bank IDS

DATASET: 1M+ Rows | Hundreds of Indicators

The Challenge

- Inefficient Wide-Format Structure
- Null Values & Naming Inconsistencies
- Overlapping Indicator Complexity

Technical Stack

Phase 1: Ingestion (Python): Reading raw CSV files using Pandas.

Phase 2: Processing (Pandas): Cleaning data and reshaping from Wide to Long format.

Phase 3: Persistence (PostgreSQL): Storing structured data with relational integrity.

Phase 4: Analytics (SQL): Running advanced queries (CTEs, Window Functions).

Phase 5: Visualization (Streamlit): Building the interactive UI.

Data Engineering (The "Reshape" Logic)

The Melt Logic

Used `pd.melt()` to collapse year columns (1970-2022) into a single "Year" attribute.

Data Cleaning

- Coerced strings to numeric and resolved NaN values.
- Pruned zero-value records for performance optimization.

Database Connection

Integrated **SQLAlchemy** for high-speed Python to PostgreSQL throughput.

Relational Database Architecture

Star Schema Design

- **Fact Table (debt_data):** Stores all numerical debt values.
- **Dimension Tables (countries_metadata, indicators_metadata):** Store descriptive data.

Constraints

Implemented Primary Keys and Foreign Keys to ensure **Referential Integrity**.

Optimization

Created SQL Views to speed up dashboard loading times.

Advanced SQL Analytics

- **Beyond Basic Queries:** The project utilizes advanced SQL techniques including:
  - **CTEs (Common Table Expressions):** For multi-step calculations.
  - **Window Functions:** Using `DENSE_RANK()` to rank countries by total debt.
  - **Aggregations:** Grouping data by indicators and regions.
- **Insight Generation:** Queries can identify countries contributing more than 5% of global debt.

The Streamlit Frontend

Multi-Tab Navigation

Organized into Global Trends, Indicators, Growth, and Data Inspector.

Dynamic Filters

Users can filter by multiple countries and specific economic indicators simultaneously.

Real-time KPIs

Automated calculation of Total Debt, Highest Debtor, and Lowest Debtor based on user selection.

Deploy

Global Trends | Indicator Breakdown | YoY Growth Velocity | Debt Composition | Detailed Data

Filter Analytics

Adjust these parameters to update all tabs dynamically.

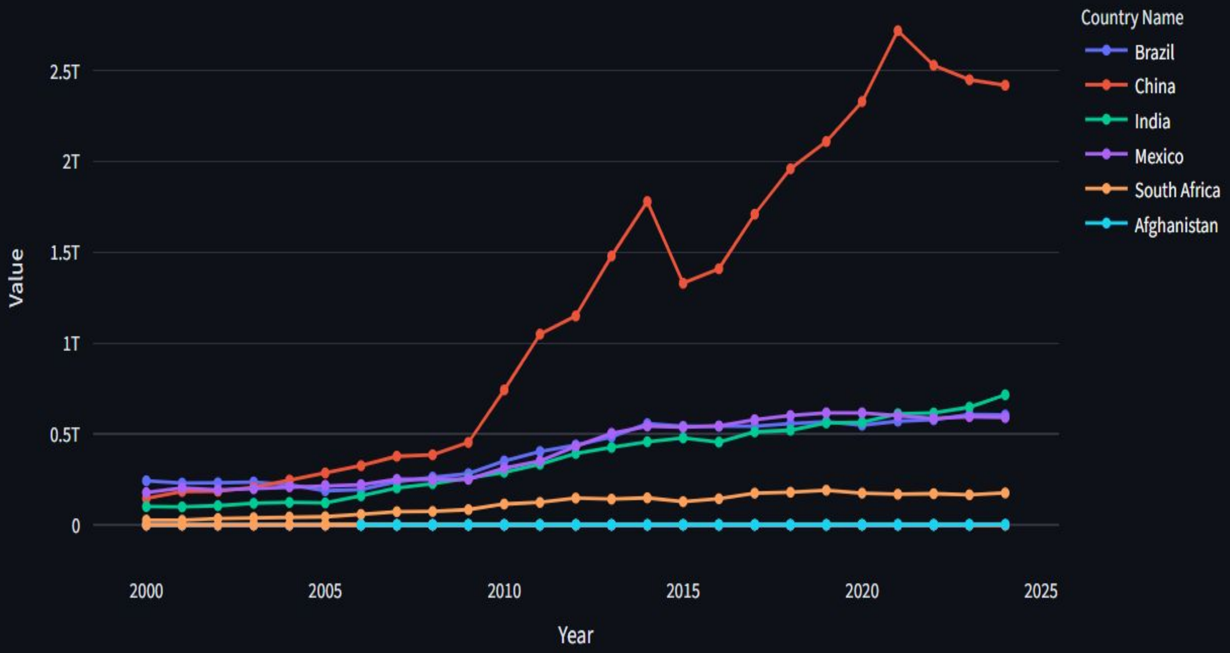
Select Countries:

India × China × Brazil × Mexico × South Africa × Afghanistan ×

Select Economic Indicators:

External debt sto… ×
Principal repay… ×
Interest paymen… ×

Historical Debt Accumulation Over Time



Advanced Insights (Tab 3 & 4)

YoY Growth Velocity

A dedicated analysis tab that calculates how fast debt is increasing year-over-year.

Debt Composition Sunburst

- A multi-level interactive chart.
- Allows users to "drill down" from a Country into its specific debt indicators.
- Shows the "Share of Parent" percentage automatically.